



AEGIS

Anonymization & Exchange Gateway
for Imaging Studies

Brain MRI

PET

CT

DICOM

AEGIS

Anonymization & Exchange Gateway for Imaging Studies

A cloud-hosted platform for secure, HIPAA-compliant sharing of medical imaging data
across all DICOM modalities

In Greek mythology, the aegis was the divine shield of Zeus and Athena — a symbol of protection. The name captures our mission: shielding patient identity while enabling the free flow of imaging data for research.

The Problem

Hospitals and research institutions need to share medical imaging data — MRI, CT, PET, ultrasound, X-ray — for multi-site clinical trials, research collaborations, and second opinions. Doing this safely is harder than it should be.

De-identification tools frequently fail. A 2015 study in *European Radiology* tested 10 free DICOM de-identification tools against 50 required PHI data elements. With default settings, only 1 of 10 tools successfully removed all required elements; 4 tools achieved success rates of 26% or less.^[1]

Metadata removal alone is not enough for brain imaging. A 2019 *New England Journal of Medicine* correspondence from Mayo Clinic showed that automated face-recognition software correctly matched de-identified brain MRI participants to their photographs in **83% of cases** — using only the scan geometry, no metadata at all.^[2] This means head imaging requires an additional step: automated defacing (removal of facial surface features from the 3D volume).

Burned-in PHI is invisible to tag-level tools. Patient names, dates of birth, and accession numbers are routinely overlaid directly on image pixels — particularly in ultrasound, CT topograms, and fluoroscopy. Standard DICOM de-identification tools operate on metadata only and miss this entirely.^[3]

Installing software at sending sites is a barrier. Platforms like XNAT require a Java-based desktop client or server daemon installed and IT-approved at every participating hospital. MIRC CTP requires a standalone Java application. Each new site adds months of IT negotiation.

Data sharing is now federally mandated — but tooling lags. The NIH Data Management and Sharing Policy (NOT-OD-21-013, effective January 25, 2023) requires all NIH-funded investigators to share scientific data as a condition of their grant award, no later than the time of first publication.^[4] Institutions are now obligated to share data they have no reliable, low-friction way to de-identify and transmit.

Result: Multi-site studies are delayed by months. In a 2021 study of global randomized trials, contract execution alone averaged 7.9 months per US site.^[5] Software installation and IT approval add to this.

Market Opportunity

The medical image exchange market is growing, driven by federal data sharing mandates and expanding multi-site clinical trial volume in oncology and neurology.

Market	2024 Estimate	2034 Projection	CAGR
Medical Image Exchange Systems ^[6]	\$3.9B	\$9.9B	9.7%

Why now:

- The NIH DMS Policy (2023) creates a compliance obligation for tens of thousands of active grants that did not previously require a data sharing plan
- A 2024 NCI-sponsored benchmark (MIDI-B, presented at MICCAI 2024) formally tested de-identification tools and found that burned-in pixel PHI and free-text fields remain the hardest unsolved problems — no existing tool handles all cases reliably^[7]
- Growing multi-site trial volume in oncology (PSMA PET, amyloid PET, whole-body MRI) and neurology (tau PET, fMRI) is driving demand for scalable data collection from distributed hospital networks
- Cloud-hosted DICOM storage (GCP Healthcare API, AWS HealthImaging) is now mature commodity infrastructure, reducing the barrier to building managed platforms

The Solution

AEGIS is a cloud-hosted platform that makes secure medical image sharing as easy as uploading a file — for any DICOM modality.

Capability	What It Does
No software to install	Upload portal runs entirely in the web browser. Hospitals use their existing computers and internet connection — no desktop app, no IT approval for new software.
Tag-level anonymization in the browser	Patient data is stripped from DICOM tags before it leaves the hospital network, following the DICOM PS3.15 Annex E Basic Confidentiality Profile — the established international standard covering all 18 HIPAA Safe Harbor identifier categories. ^[8]
Automated defacing for head	MRI, CT, and PET studies of the head and brain undergo server-side defacing (removal of facial surface geometry) before

scans	entering the research archive. Non-head studies bypass this step automatically.
Burned-in PHI detection	Automated OCR scans image pixels for overlaid text (patient names, dates, accession numbers) that tag-level tools miss. Studies with detected pixel PHI are flagged for review.
Supports all DICOM modalities	MRI, CT, PET, PET/CT, ultrasound, X-ray, mammography, nuclear medicine, and more — using the same upload and anonymization workflow.
Centralized audit trail	Every upload, approval, rejection, and data export is logged with a timestamp and actor. Institutions can demonstrate HIPAA compliance from a single dashboard.
Clinician review before release	Administrators review anonymized images in a web-based DICOM viewer (OHIF) before approving studies for sharing. Nothing is shared automatically without human sign-off.

How De-identification Works

AEGIS follows a two-phase approach that addresses the limitations documented in the literature:

Phase 1 – In the browser (before upload):

- └─ DICOM tags stripped per PS3.15 Annex E Basic Profile [8]
- └─ 18 HIPAA Safe Harbor identifiers addressed [9]
- └─ Reviewer sees before/after tag comparison before confirming
- └─ Only de-identified data is transmitted – PHI never leaves the hospital

Phase 2 – On the server (after upload):

- └─ De-identification completeness check
- └─ OCR scan of image pixels for burned-in text [3]
- └─ Head imaging → automated defacing [2]
- └─ Administrator review and approval before data is shared

This two-phase design directly addresses the gaps identified in the Aryanto (2015) and Schwarz (2019) studies.

Supported Modalities

Category	Modalities
Cross-sectional	MRI, CT, PET, PET/CT, PET/MR
Projection	X-Ray (CR/DR), Mammography (FFDM/DBT), Fluoroscopy
Nuclear Medicine	SPECT, Planar Scintigraphy
Ultrasound	Echocardiography, Abdominal, Vascular, OB/GYN
Radiation Therapy	RT Structure, RT Plan, RT Dose, RT Image
Other DICOM	Secondary Capture, Structured Reports, Key Images

Initial focus: Brain MRI, PET, and CT — these exercise the full pipeline including defacing. All other modalities use the same upload and de-identification workflow.

Technology

Layer	Technology	Notes
Backend	Go on Google Cloud Run	Compiled binary, minimal dependencies, fast cold starts
Frontend	React + TypeScript	Runs in any modern browser, no installation required
Database	PostgreSQL 15 (Cloud SQL)	Audit trail, project/user management, routing rules
DICOM Storage	GCP Healthcare API	Managed DICOMweb storage, no servers to maintain
Analytics	BigQuery	DICOM metadata export, compliance reporting
OCR / PHI Detection	Google Cloud Document AI	Detects burned-in text in image pixels

Defacing	mri_deface (FreeSurfer)	Established tool, used in neuroimaging research
DICOM Viewer	OHIF Viewer (v3)	Open-source, browser-based, supports all modalities
Infrastructure	Terraform	Reproducible, version-controlled infrastructure

On the use of automated tools: AEGIS uses automated tools to assist with — not replace — human review. Automated de-identification flags potential issues; a trained administrator reviews and approves every study before it is shared. Automated defacing quality is reviewed side-by-side against the original in the admin interface.

Compliance Framework

AEGIS is designed around established regulatory and technical standards:

Standard	Reference	What It Covers
HIPAA Privacy Rule	45 CFR § 164.514(b)	De-identification methods: Safe Harbor (18 identifiers) and Expert Determination
DICOM Confidentiality Profile	NEMA PS3.15 Annex E	Tag-level de-identification actions for all DICOM attributes
NIH Data Management Policy	NOT-OD-21-013 (eff. Jan 25, 2023)	Data sharing obligations for NIH-funded investigators
GCP Security	BAA, VPC, CMEK	Business Associate Agreement, encrypted storage, network controls

AEGIS does not create a new de-identification standard — it implements the ones that already exist.

Competitive Landscape

	AEGIS	XNAT	Flywheel	MIRC CTP
Hosting	Cloud-hosted (GCP)	Self-hosted	Commercial SaaS	On-premises
Install at sending site	None — browser only	Desktop Java client	CLI tool	Java application
Modalities	All DICOM	Neuroimaging focus	All imaging	All imaging
Defacing	Automated, reviewable	Manual or plugin	Built-in	None
Open source	Yes	Yes	No	Yes
Audit trail	Centralized, per-study	Per-instance	Built-in	Limited

Key differences from existing platforms:

1. **No software installation at sending sites** — the full anonymization workflow runs in the browser
2. **Two-phase de-identification** — both tag-level and pixel-level (burned-in PHI) are addressed
3. **Defacing is automated and reviewable** — not optional or manual
4. **Not specialty-specific** — designed for all DICOM modalities from the start

Phased Roadmap

Phase 1 — Foundation (Complete)

- Cloud infrastructure (Terraform, GCP)
- Browser-based upload portal with DICOM tag anonymization and before/after preview
- Go API with DICOM ingest and PostgreSQL audit trail

- Admin dashboard with OHIF viewer for QC, approve/reject workflow

Phase 2 — Defacing + Notifications (Complete)

- Automated mri_deface pipeline on Cloud Run for head imaging
- Side-by-side before/after defacing review in admin interface
- Email notifications for upload confirmation, approval, and rejection

Phase 3 — Operations (Complete)

- Multi-project routing engine with configurable rules by modality, source, and anatomy
- Institution management with project-scoped roles
- Configurable anonymization profiles per project (which tags to retain for research)
- Audit log viewer with filtering
- Admin user management and access control

Phase 4 — Advanced Validation (Next)

- Burned-in PHI detection using OCR on image pixels (addresses gap identified in [3, 7])
- Automated image quality assessment (motion artifact detection, coverage completeness)
- BIDS format conversion for neuroimaging research output
- Batch import tools for historical data migration

Cost Estimate (Development / Proof of Concept)

Resource	Monthly
Cloud Run (API + dashboard)	~\$5–15 (covered by free tier in dev)
Cloud SQL (PostgreSQL, small instance)	~\$10
Healthcare API (DICOM store)	~\$0.50/GB stored
Cloud Storage (staging)	~\$0.02/GB
BigQuery	Free tier (1 TB queries/month)

Document AI (OCR for burned-in PHI)	Pay per page (minimal during dev)
Total (development)	~\$20–40/month

Production costs scale with data volume. A 1,000-session multi-site study (~500 GB) would cost approximately \$50–100/month in storage.

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AEGIS

Secure, browser-based medical image sharing for any DICOM modality — built on established de-identification standards, with automated defacing, pixel-level PHI detection, and administrator review before any data is released.